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Proof of Concept
GUJCOST-ROBOFEST 5.0

Autonomous Drone Swarm for Minefield Detection and Safe Human Navigation Using Gesture-Based Control

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System Overview

The proposed system is a **multi-drone aerial robotic platform** designed for safe minefield mapping and human navigation through coordinated autonomous operation. The system comprises multiple quadrotor drones that work collaboratively to survey the operational area, detect mine-like objects, and compute a safe traversal path. The system detects simulated surface-level mine markers as defined in the ROBOFEST 5.0 guidelines to validate swarm coordination and safe path planning.

Each drone is built on a **customized quadrotor frame** optimized for payload integration, vibration reduction, and flight stability. The mechanical design ensures proper center-of-mass alignment, enabling stable low-altitude flight required for visual detection tasks.

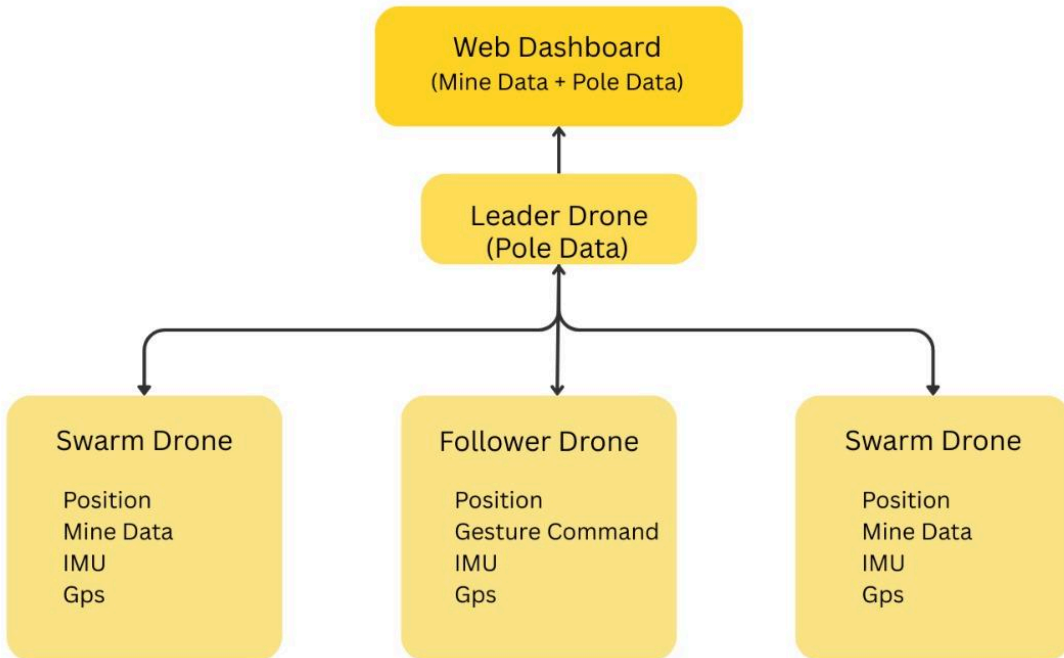
All drones are equipped with a **Mini Pix flight controller**, responsible for low-level flight stabilization and control. A **heterogeneous onboard computing architecture** is adopted across the fleet. The **Leader Drone** is equipped with a **Raspberry Pi 5**, which handles computationally intensive tasks such as centralized data fusion, AI-based gesture recognition, and global path planning. The **Follower and Swarming Drones** are equipped with **Raspberry Pi 4**, which perform lightweight perception, contour-based detection, and communication tasks.

The propulsion system consists of **brushless DC motors** driven by **Cyclone 45 A electronic speed controllers (ESCs)**, ensuring smooth thrust response and stable maneuvering. During mine detection, the drones operate at a low altitude of approximately **2 meters**, where the **Raspberry Pi Camera Module 3** provides sufficient ground coverage and resolution for accurate visual detection.

A **role-based operational architecture** is implemented:

- The **Follower Drone** performs initial boundary mapping.
- The **Leader Drone** coordinates mission planning and global decision-making.
- The **Swarming Drones** execute systematic low-altitude scanning.

To ensure accurate localization, each drone employs an **Extended Kalman Filter (EKF)-based IMU–GPS sensor fusion algorithm**, enabling reliable coordinate tagging and consistent global mapping.



High-Level System Interaction Diagram

1. Steps in the Working of the Proof of Concept (POC)

1.1 System Initialization and Mission Setup

All drones are placed in the designated start zone with propellers disarmed. Upon powering ON, each drone performs an internal system health check, during which onboard sensors such as the IMU and GPS undergo calibration to ensure reliable state estimation. Communication links between all drones and the ground control station are established and verified to confirm network availability and system readiness.

The **Follower Drone** is initialized first. After achieving a stable GPS lock, it takes off and performs an initial boundary reconnaissance flight. During this phase, the Follower Drone records the GPS coordinates of the four boundary corners of the operational area. These boundary coordinates are transmitted to the **Leader Drone**, which uses them to define and lock the mission workspace.

Next, the **Leader Drone** is armed and processes the received boundary information. Based on the defined mission area, it partitions the operational space into multiple sub-regions corresponding to the number of available Swarming Drones. The Leader Drone then ascends to a higher altitude to maintain global situational awareness and to coordinate mission planning, task allocation, and inter-drone communication.

Finally, the **Swarming Drones** are initialized sequentially. Each Swarming Drone receives an assigned sub-region and begins low-altitude scanning operations within its designated area. During mission execution, the Leader Drone continuously monitors the progress and operational status of each Swarming Drone. If a drone completes its assigned region early or encounters slower scanning due to dense detections or operational constraints, the Leader Drone dynamically reallocates remaining unscanned or partially scanned regions to other available Swarming Drones. This dynamic coverage reallocation ensures balanced workload distribution, efficient area coverage, and reduced overall mission time.

1.2 Localization and Sensor Fusion

Each drone employs an onboard localization framework based on sensor fusion to obtain stable and drift-corrected state estimates. The Inertial Measurement Unit (IMU) provides high-frequency orientation and motion data, while the GPS module supplies absolute global positional information. During low-altitude flight, altitude estimation is enhanced using LiDAR measurements to improve relative spatial awareness and terrain awareness.

An Extended Kalman Filter (EKF) is executed locally on each drone to fuse IMU, GPS, and altitude data, generating a reliable estimate of position, velocity, and orientation while significantly reducing sensor noise and drift. This fused state estimate enables approximate hazard localization of detected simulated mine markers and ensures consistent relative mapping with sufficient accuracy to maintain the mandatory 1-meter safety buffer across all drones operating within the swarm.

1.3 Real-Time Distributed Communication

Inter-drone communication is implemented using a **Wi-Fi–based UDP protocol**, operating at approximately 5–10 Hz to ensure low-latency data exchange during mission execution. All drones communicate using lightweight JSON-formatted packets, minimizing communication overhead while maintaining timely synchronization across the fleet.

The communication framework follows a distributed sensing with centralized coordination model. Each drone performs local perception and state estimation while exchanging only task-relevant information with the Leader Drone, reducing bandwidth usage and improving scalability.

1.3.1 Communication from Swarming Drones

The Swarming Drones are responsible for transmitting sensing and detection-related information to the Leader Drone. Each Swarming Drone periodically broadcasts the following data:

- Current GPS position and EKF-estimated pose
- Validated mine detection coordinates
- Detection confidence and status information
- Operational state and heartbeat signals

This transmitted data enables the Leader Drone to aggregate detection results, eliminate redundant detections, and update the global mine map in real time. Additionally, shared positional information allows the Leader Drone to maintain safe inter-drone separation and coordinated area coverage, preventing overlap and collision during dense scanning operations.

1.3.2 Communication with the Follower Drone

The Follower Drone receives navigation commands and path updates from the Leader Drone in real time. These commands include:

- Waypoint coordinates along the computed safe path
- Motion directives for speed and heading adjustments
- Update signals in response to newly detected hazards

The Follower Drone continuously transmits its position and operational status back to the Leader Drone. This bidirectional communication enables closed-loop monitoring and allows the Leader Drone to issue timely path corrections in response to localization deviations or newly identified hazards.

1.3.3 Communication Reliability and Synchronization

To maintain synchronized operation, all drones operate at a fixed communication rate of approximately 5–10 Hz. The use of UDP ensures minimal latency, while periodic heartbeat and status messages allow the Leader Drone to monitor the health and availability of each drone.

In the event of temporary packet loss or communication degradation, each drone continues operation using the most recent valid command and state information. Upon restoration of communication, the system automatically resynchronizes with the Leader Drone, ensuring robust and uninterrupted mission execution.

1.4 Data Acquisition and Mine Detection

Mine detection in the Proof of Concept stage is implemented using a **vision-based contour detection methodology**, where mine-like objects are simulated using visually distinguishable markers placed on the ground. The objective of this approach is to validate detection, localization, and coordination logic rather than real explosive sensing.

1.4.1 Image Acquisition and Preprocessing

Each Swarming Drone is equipped with a downward-facing camera that continuously captures video frames during low-altitude flight at approximately **2 meters**. The captured frames are converted to an appropriate color space and undergo preprocessing steps, such as noise reduction, smoothing, and thresholding, to enhance the contrast between mine-like objects and the background. These preprocessing steps ensure consistent contour extraction under varying lighting and surface conditions.

1.4.2 Contour Extraction and Geometric Filtering

After preprocessing, contours are extracted from the binary image using standard contour detection algorithms. Each detected contour is evaluated using geometric constraints, including:

- Minimum and maximum contour area
- Shape consistency and aspect ratio
- Contour solidity and boundary smoothness

Contours that do not satisfy these constraints are discarded as false positives. This geometric filtering stage ensures that only objects resembling predefined mine-like characteristics are considered for further analysis.

1.4.3 Temporal Validation and Confidence Assignment

To further improve detection robustness, **temporal consistency checks** are applied. A detected contour is confirmed as a valid mine-like object only if it appears consistently across **multiple consecutive frames** while the drone undergoes slight motion.

A confidence score is assigned to each confirmed detection based on:

- Geometric match quality
- Number of consecutive frames detected
- Stability of contour position across frames

Only detections exceeding a predefined confidence threshold are forwarded for mapping and planning.

1.4.4 Georeferencing and Data Transmission

Once a mine-like object is confirmed, its image-frame coordinates are transformed into **world coordinates** using EKF-based GPS position, IMU orientation, and altitude information. This transformation enables accurate georeferencing of detected hazards within the global map.

The georeferenced mine coordinates, along with associated confidence information, are transmitted to the **Leader Drone** via low-latency UDP communication. The Leader Drone aggregates detections from all Swarming Drones, eliminates redundancy, and updates the centralized mine map used for safe path planning.

1.5 Centralized Data Fusion and Path Planning

The **Leader Drone** acts as the central decision-making unit. It aggregates detection data received from all Swarming Drones. Redundant detections are filtered, and confirmed mine locations are added to a centralized global map.

Using this map, the Leader Drone computes a **safe traversal path** using a graph-based path planning algorithm while maintaining a safety buffer around detected hazards. The computed path is continuously updated if new hazards are detected or if localization deviations occur.

1.6 Human Guidance and Closed-Loop Feedback

Human guidance in the proposed system is achieved through a combination of gesture-based command interpretation and closed-loop feedback control, enabling intuitive interaction between the human participant and the drone swarm.

All drones are equipped with onboard cameras to support visual perception and situational awareness. However, gesture recognition and command interpretation are executed exclusively on the Leader Drone, which is equipped with a higher computational capability.

1.6.1 Hand Gesture Capture and Interpretation

The Leader Drone uses its onboard camera to capture visual input of the human participant. Predefined hand gestures are used to represent high-level navigation intents such as direction selection, pause, or resume movement. The captured video frames are processed on the Leader Drone using a lightweight gesture recognition pipeline.

At the Proof of Concept stage, gesture recognition is implemented using a combination of image preprocessing and feature-based classification. This enables reliable detection of predefined gestures under controlled conditions without requiring extensive training data. The recognized gesture is translated into a high-level command, which influences the path selection or movement strategy generated by the Leader Drone.

1.6.2 Path Execution and Closed Loop Tracking

Based on the interpreted human intent and the computed safe path, the Leader Drone generates waypoint-based navigation commands. These commands are transmitted to the Follower Drone, which acts as the physical guide for the human participant.

The Follower Drone continuously tracks its position using EKF-based localization and monitors deviations from the planned path. Any deviation beyond acceptable thresholds is corrected in real time through waypoint updates and motion adjustments.

1.6.3 Closed-Loop Feedback and Dynamic Replanning

The entire system operates within a closed-loop feedback architecture. Sensor data, detection results, and positional information from all drones are continuously shared with the Leader Drone.

If a new mine-like object or hazard is detected during traversal, the Leader Drone immediately updates the global map and recomputes the safe path. Updated navigation commands are then sent to the Follower Drone, ensuring adaptive and safe guidance throughout the mission.

This feedback-driven approach enables the system to respond dynamically to environmental changes while maintaining continuous human guidance.

Electronic Description

The electronic system of the proposed robot is designed to support fully onboard autonomous operation without reliance on any ground control station. Each drone is built around a Mini Pix flight controller, which is responsible for low-level flight stabilization and attitude control. The flight controller continuously receives motion data from the onboard IMU and position data from the GPS module to maintain stable flight.

Each drone integrates a companion computer based on its assigned role. The Leader Drone is equipped with a Raspberry Pi 5 for centralized processing, while the Follower and Swarming Drones use Raspberry Pi 4 for distributed perception, communication, and navigation tasks.

The Raspberry Pi Camera Module 3, mounted in a downward-facing configuration, captures ground images during low-altitude flight. These images are processed on the companion computer to detect mine-like objects. Detected pixel coordinates are combined with EKF-refined position and orientation data to estimate the approximate global position of detected mines.

High-level navigation and coordination commands generated by the companion computer are transmitted to the Mini Pix flight controller using a serial communication interface. The flight controller converts these commands into motor control signals and sends pulse-width modulation (PWM) outputs to the Cyclone 45 A electronic speed controllers, which regulate the speed of the brushless DC motors. The motors dynamically adjust thrust to maintain stability and follow the assigned trajectories.

Inter-drone communication is handled through onboard communication modules, enabling the exchange of processed mapping data and navigation commands between the Leader Drone, Swarming Drones, and Follower Drone. In addition, mission data such as detected mine locations, drone positions, and safe paths is transmitted to a web-based dashboard for real-time visualization and monitoring.

The power distribution system ensures a regulated voltage supply to all electronic components, while onboard regulators and filtering elements maintain stable operation during dynamic load conditions.

2. List of Components Used in POC Stage

A. ELECTRONIC COMPONENTS

Component	Function in POC	Justification
Raspberry Pi 5 (16GB)	High-performance processing for AI, Fusion, and Localization Logic.	Uniform, high-memory capacity ensures low-latency and high-throughput processing for centralized data fusion and complex A* path planning.
Raspberry Pi 4	Capture downward frames, detect mines via contours, filter geometrically (area, shape, aspect ratio), and transmit coordinates to Leader Drone via UDP.	Provides sufficient processing capability for real-time image processing and local EKF-based state estimation, while reducing overall system cost and power consumption.
Mini Pix Flight Controller (Radio Link)	Provides low-level flight stabilization and high-quality **IMU** data.	Established, reliable platform running ArduPilot/PX4 firmware, ensuring predictable flight control and clean sensor data for the companion computer.
Cyclone 45A ESC	Provides robust and precise motor control (45A rating).	A high current rating ensures reliable power management for high-thrust motors, which is crucial for responsive maneuvers and stability under load.
Velox 1750kv Motor	Provides propulsion (1750kv) for high-efficiency flight.	Selected for optimal kV rating, balancing thrust and endurance required for the 15+ minute mission time.

UBEC (Voltage Regulator)	Regulates power supply from the main battery for sensitive electronics.	Provides clean and stable 5V/12V power to companion computers and onboard cameras, preventing voltage fluctuations and brownouts during high-load conditions.
12 MP AI Camera	Provides high-resolution input for **Real-Time Gesture Hand Control**.	High-resolution (12 MP) and wide field-of-view cameras are required for accurate and responsive detection of human gestures at varying distances.
RPi Camera Module 3	Provides vision input for mine detection.	Optimized for use with Raspberry Pi, delivering sufficient quality for color-based object detection during the 2m altitude sweep.
HGLRC M100 Pro GPS	Provides absolute global position data for georeferencing.	Multi-constellation GNSS provides robust global position locking, serving as the essential absolute correction input for the fusion logic.

B. Structural Components

- Custom Carbon Fibre Frame (300x300mm): Lightweight and durable frame for all four drones, balancing endurance with structural integrity.
- Gemfan 51499 Hurricane PC 3 Blade Prop | 16 (4 sets) | All Drones | Propeller optimized for the Velox 1750kv motors, ensuring high thrust efficiency and responsive handling. |
- 3D Printed Parts: Custom mounts for the Raspberry Pi 5s and all peripheral electronics, ensuring precise and secure component placement.

3. Software Used & Developed

This section describes the software stack used to implement perception, communication, coordination, and control in the proposed multi-drone system. The software architecture follows a distributed sensing and centralized decision-making model, with task allocation based on drone roles.

3.1 Software Stack Overview

Software / Framework	Purpose	Usage in System
ROS (Robot Operating System)	Middleware for inter-process communication	Used on all drones for modular sensor handling, message passing, and task coordination.
Python	Primary programming language	Used for onboard perception, sensor fusion, communication handling, path-planning logic, and backend API development.
OpenCV	Computer vision library	Performs image preprocessing, contour extraction, geometric filtering, and temporal validation for simulated surface-level mine marker detection.
MediaPipe	Vision-based hand tracking framework	Captures hand landmarks on the Leader Drone for gesture-based high-level command recognition under controlled conditions.
TensorFlow Lite	Lightweight ML inference framework	Executes optimized gesture recognition models on the Leader Drone for real-time onboard inference.
PyTorch	Model training and prototyping	Used offline for training and prototyping gesture recognition models; not used during onboard flight execution.
MAVLink	Flight controller communication protocol	Enables reliable command and telemetry exchange between the companion computer and the Mini Pix flight controller.

FastAPI (Backend)	High-performance backend framework	Receives telemetry, detection data, and path updates from the drone swarm, manages mission-level data, and exposes REST/WebSocket APIs for real-time visualization.
React.js (Frontend)	Web-based user interface	Implements an interactive dashboard to visualize drone positions, detected simulated mine markers, safe paths, and overall mission status in real time.
Leaflet.js	Lightweight mapping library	Renders the minefield map and drone trajectories in a browser-based environment without reliance on external map services.
SVG Overlays	Vector-based visualization	Used with Leaflet to draw detected simulated mine markers, 1-meter exclusion zones, and dynamically updated safe path corridors with high visual clarity.

3.2 Role-Based Software Allocation

3.2.1 Leader Drone (Raspberry Pi 5)

The Leader Drone executes computationally intensive and centralized tasks, including:

- Sensor data aggregation from all drones
- Global map generation and maintenance
- Gesture-based human command interpretation using MediaPipe + TensorFlow Lite
- Centralized path planning using the A* algorithm
- Mission coordination and command distribution

The higher computational capacity of Raspberry Pi 5 enables real-time AI inference and planning without affecting system responsiveness.

3.2.2 Swarming Drones (Raspberry Pi 4)

The Swarming Drones perform distributed sensing and local processing tasks:

- Image acquisition from a downward-facing camera
- Preprocessing and contour-based mine detection using **OpenCV**
- Temporal validation of detected objects
- Conversion of image coordinates to world coordinates using EKF outputs
- Transmission of validated detection data to Leader Drone

3.2.3 Follower Drone (Raspberry Pi 4)

The Follower Drone executes navigation and guidance tasks:

- Reception of waypoint-based navigation commands from Leader Drone
- Local position monitoring using EKF-based localization
- Execution of motion commands via the flight controller
- Continuous status and position feedback to the Leader Drone

3.3 Communication Architecture

The system employs a **hybrid communication architecture**, combining wired and wireless links to ensure reliable and low-latency operation.

3.3.1 Companion Computer to Flight Controller Communication

Communication between the **Raspberry Pi (companion computer)** and the **Mini Pix flight controller** is established via a **USB serial connection** using the **MAVLink protocol**.

Functionality includes:

- Transmission of velocity, position, and waypoint commands
- Reception of telemetry data (IMU, GPS, attitude)
- Arming, mode switching, and safety monitoring

This wired communication ensures deterministic and reliable command execution with minimal latency.

3.3.2 Inter-Drone Communication

Inter-drone communication is implemented using a **Wi-Fi–based UDP protocol**, enabling low-latency data exchange at approximately **5–10 Hz**.

Data packets are transmitted in **lightweight JSON format**, reducing overhead and enabling rapid synchronization across the drone fleet.

Data Exchanged	Source	Destination
Mine coordinates & confidence	Swarming Drones	Leader Drone
Global map & safe path	Leader Drone	Follower Drone

Position & status updates	All Drones	Leader Drone

3.3.3 Communication Robustness

UDP is selected to minimize latency and support real-time coordination. Periodic heartbeat and status messages allow the Leader Drone to monitor drone availability. In the event of temporary packet loss, the system continues to operate using the most recent valid data, ensuring mission continuity.

3.4 Path Planning Software

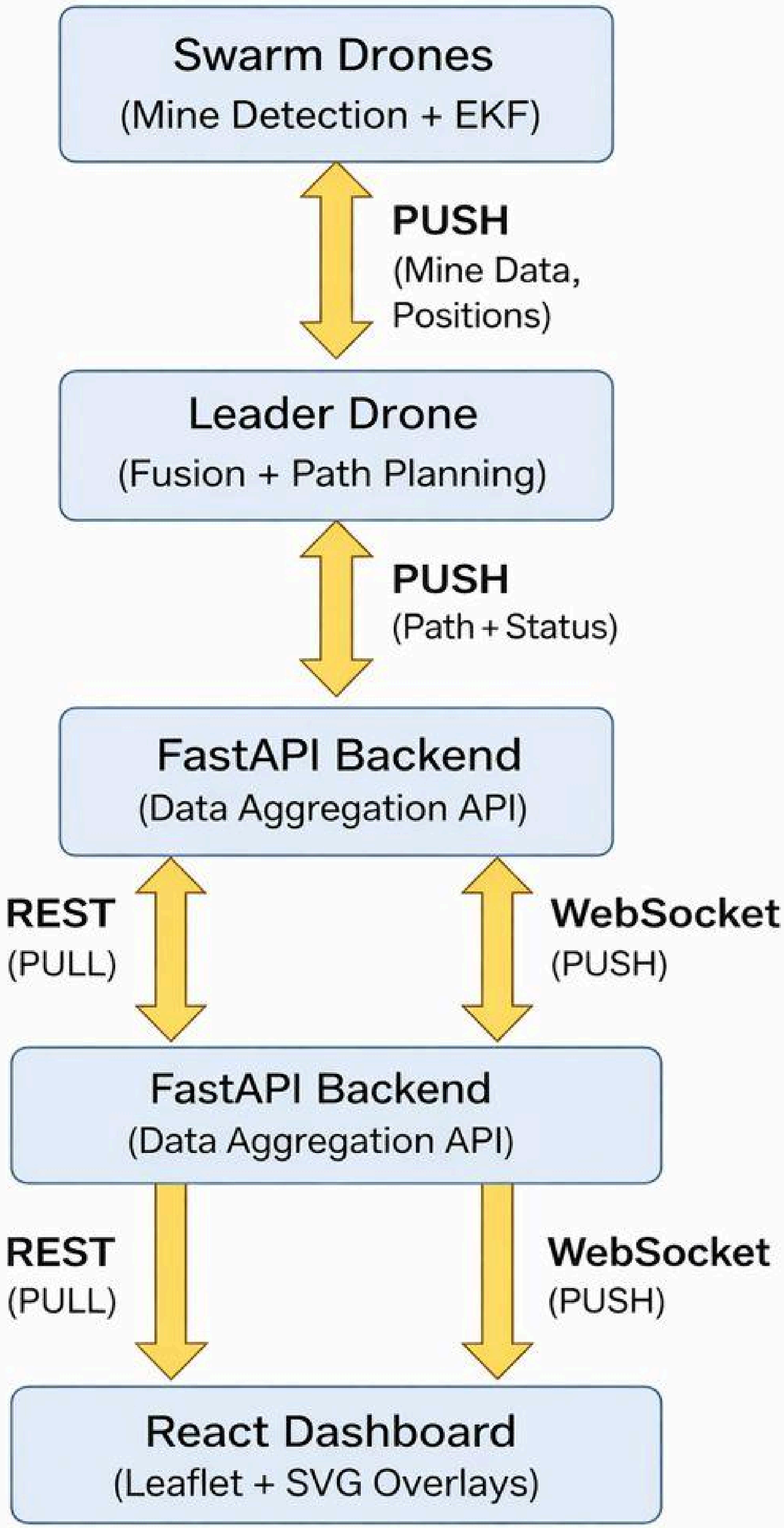
Safe path planning is implemented on the Leader Drone using an **iterative A*** search algorithm. The algorithm operates on the centralized global map generated from swarm detections and computes an optimal traversal path while maintaining a predefined safety buffer around detected mine-like objects.

Path updates are triggered dynamically when new hazards are detected, ensuring adaptive navigation.

3.5 Software Design Justification

A distributed software architecture with centralized planning is adopted to balance computational load, power consumption, and scalability. Lightweight perception and communication tasks are assigned to Swarming and Follower Drones, while AI-based inference and global decision-making are executed exclusively on the Leader Drone.

This design ensures efficient utilization of onboard resources while maintaining synchronized and reliable multi-drone operation.



Dashboard Data Flow

4. Features of the Multi-Agent System

1. **Uniform High-Performance Processing:** Utilizing a heterogeneous computing architecture with Raspberry Pi 5 on the Leader Drone and Raspberry Pi 4 on Swarming and Follower Drones ensures efficient load distribution, low-latency coordination, and optimized power consumption across the swarm.
2. **AI Gesture Control:** Direct, intuitive human control of the Follower Drone via AI-processed hand movements, eliminating the need for a separate joystick controller.
3. **Coordinated Dual-Sweep Mapping:** Simultaneous mine detection by two dedicated Swarming Drones ensures high area coverage efficiency.
4. **Autonomous Dynamic Pathfinding:** The Iterative A* algorithm provides instantaneous recalculation of the safest route as new mines are detected.
5. **Robust Localization:** Sensor fusion of GPS, IMU, and LiDAR provides stable positional data for high-accuracy mapping.

5. Concurrence / Deviation from Stage 1 Report (Ideation)

This section confirms the successful design and modeling of the proposed architecture and clarifies key technical choices required for autonomous operation.

Deviation/Concurrence from Ideation	Justification	POC Validation Outcome
Core Processor	**DEVIATION:** Leader drones are upgraded to **Raspberry Pi 5 (16GB)**. Other follower and swarming drones uses Raspberry Pi 4.	**Confirmed.** Necessary for uniform, high-speed processing, eliminating bottlenecks in sensor fusion and AI latency.
Flight Controller	**DEVIATION:** Switched to **mini Pix FC** and **Cyclone 45A ESC**.	**Confirmed.** Provides established ArduPilot/PX4 firmware compatibility and robust power handling for superior flight stability.

Gesture Hand Control	**CONCURRENCE:** Implements the Human Guidance goal via AI vision processing on the Leader Drone.	**Confirmed.** AI/CV capability for human control is integrated into the hardware design.

6. Next Steps (Level 3 Prototype)

The Level 3 prototype phase will focus on **system integration, robustness, and capability enhancement**, transitioning from POC validation to a fully functional and demonstrable multi-drone system.

6.1 Advanced Gesture Recognition Pipeline

Design, training, and optimization of a **lightweight machine learning pipeline** for real-time hand-gesture command detection on the **Leader Drone (Raspberry Pi 5)** using **MediaPipe, OpenCV, and TensorFlow Lite**. The focus will be on improving gesture recognition accuracy, latency, and robustness under varying lighting and background conditions.

6.2 Hardware-in-the-Loop (HIL) Testing

Integration of localization and perception logic with physical drone sensors to perform **Hardware-in-the-Loop testing**. This phase will involve real flight experiments to tune EKF fusion parameters and control gains, ensuring improved positional accuracy and flight stability in real-world conditions.

6.3 Swarm Communication and Stress Testing

Conduct full-range operational tests (up to **100 meters**) to evaluate the reliability of **UDP-based inter-drone communication**. This includes validating data throughput, latency, packet loss tolerance, and synchronization across all four drones under dynamic mission conditions.

6.4 Mine Detection Pipeline Finalization

Complete integration of the **vision-based contour detection pipeline** on the Swarming Drones using OpenCV. Detection accuracy will be validated through controlled experiments, and confidence thresholds will be tuned to minimize false positives while maintaining reliable hazard detection.

6.5 Mechanical Integration and Final Assembly

Finalize the custom drone frame assembly and complete full integration of all onboard electronics, sensors, power systems, and communication modules. Mechanical refinements will focus on vibration isolation, camera alignment, and thermal management to improve perception performance.

6.6 Thermal-Based Detection for Subsurface Mine Identification (Future Enhancement)

As a Level-3 enhancement, the system proposes the integration of **thermal imaging** to enable the detection of **subsurface or partially buried mines** that are not reliably observable using RGB vision alone.

Buried objects often alter the **thermal behavior of the surrounding soil** due to differences in thermal conductivity, heat capacity, and emissivity when compared to natural terrain. During thermal transitions (e.g., early morning heating or evening cooling), these differences can manifest as localized temperature anomalies on the ground surface. Thermal cameras are capable of capturing such variations, even when the object itself is not visually exposed.

In this approach, a **lightweight thermal camera** mounted on a **Swarming Drone** will periodically acquire **thermal maps** of the operational area. These maps represent spatial temperature distributions and are analyzed to identify abnormal thermal patterns that may indicate the presence of buried objects.

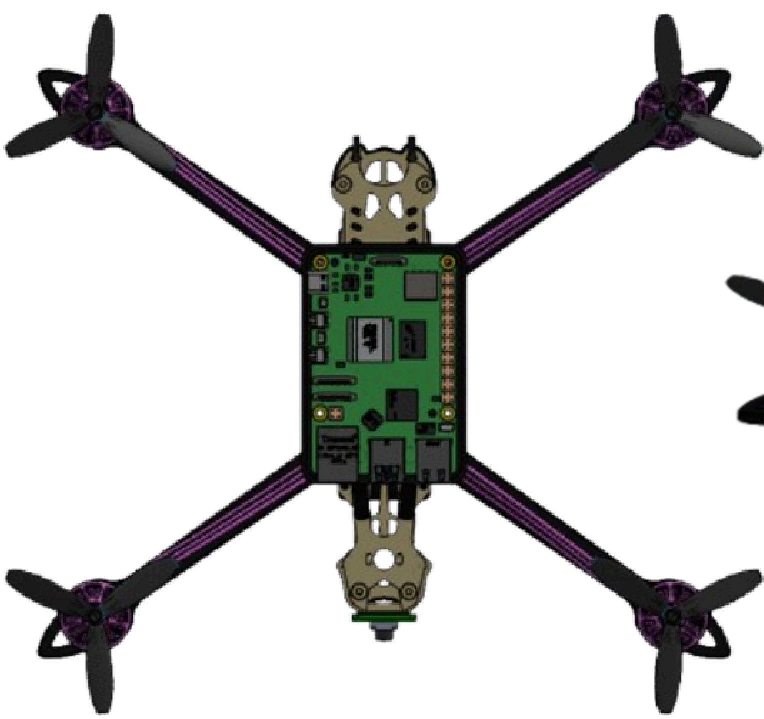
To improve reliability, thermal detections will not be used in isolation. Instead, **multi-modal sensor fusion** will be applied by correlating:

- Thermal anomalies from the infrared camera
- Surface-level detections from RGB vision (shape, texture, and contextual cues)
- Geospatial consistency across multiple drone passes

Regions where both thermal irregularities and visual indicators overlap are assigned a **higher detection confidence score**, while isolated detections are filtered to reduce false positives caused by environmental factors such as rocks, vegetation, or uneven solar exposure.

This fused RGB–thermal detection pipeline extends the system's capability beyond surface-visible mines, enhances robustness under challenging lighting conditions, and provides a scalable foundation for advanced subsurface threat identification in future deployments.

7. Mechanical Design and CAD Model



8. PHOTOS

9. Links for Video and Code.

https://drive.google.com/drive/folders/1Zn6hnmbidfBRnTnsvFy0fLmwMWv67bxV?usp=sharing

https://github.com/P07awan/Robofest-5.0